

Session-13

1. Java Priority Queue:

```
import java.util.ArrayList;
import java.util.List;
import java.util.Scanner;
import java.util.ArrayList;
import java.util.Comparator;
import java.util.List;
import java.util.PriorityQueue;

class Student {
    private final int id;
    private final String name;
    private final double cgpa;

    public Student(int id, String name, double cgpa) {
        this.id = id;
        this.name = name;
        this.cgpa = cgpa;
    }

    public int getID() {
        return id;
    }

    public String getName() {
        return name;
    }

    public double getCGPA() {
        return cgpa;
    }
}

class Priorities {
    public List<Student> getStudents(List<String> events) {
        PriorityQueue<Student> pq = new PriorityQueue<>(
            Comparator.comparing(Student::getCGPA).reversed()
                .thenComparing(Student::getName)
                .thenComparing(Student::getID)
        );

        for (String event : events) {
            String[] parts = event.split(" ");
            if (parts[0].equals("ENTER")) {
                String name = parts[1];
                double cgpa = Double.parseDouble(parts[2]);
                int id = Integer.parseInt(parts[3]);
                pq.add(new Student(id, name, cgpa));
            }
        }
    }
}
```

```

        } else if (parts[0].equals("SERVED")) {
            pq.poll();
        }
    }

    List<Student> remainingStudents = new ArrayList<>();
    while (!pq.isEmpty()) {
        remainingStudents.add(pq.poll());
    }

    return remainingStudents;
}

}

public class Solution {
    private final static Scanner scan = new Scanner(System.in);
    private final static Priorities priorities = new Priorities();

    public static void main(String[] args) {
        int totalEvents = Integer.parseInt(scan.nextLine());
        List<String> events = new ArrayList<>();

        while (totalEvents-- != 0) {
            String event = scan.nextLine();
            events.add(event);
        }

        List<Student> students = priorities.getStudents(events);

        if (students.isEmpty()) {
            System.out.println("EMPTY");
        } else {
            for (Student st: students) {
                System.out.println(st.getName());
            }
        }
    }
}

```

Output:



2. Java ArrayList:

```
import java.io.*;
import java.util.*;

public class Solution {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int n = scanner.nextInt();
        ArrayList<ArrayList<Integer>> lines = new ArrayList<>();

        for (int i = 0; i < n; i++) {
            int d = scanner.nextInt();
            ArrayList<Integer> line = new ArrayList<>();
            for (int j = 0; j < d; j++) {
                line.add(scanner.nextInt());
            }
            lines.add(line);
        }

        int q = scanner.nextInt();
        for (int i = 0; i < q; i++) {
            int x = scanner.nextInt();
            int y = scanner.nextInt();

            try {
                System.out.println(lines.get(x - 1).get(y - 1));
            } catch (IndexOutOfBoundsException e) {
                System.out.println("ERROR!");
            }
        }

        scanner.close();
    }
}
```

}

}

Output:

✓ Test case 0

✓ Test case 1

✓ Test case 2 

✓ Test case 3 

✓ Test case 4 

✓ Test case 5 

Compiler Message

Success

Input (stdin)

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1 5

2 5 41 77 74 22 44

3 1 12

4 4 37 34 36 52

5 0

6 3 20 22 33

7 5

8 1 3